

ASSIGNMENT 1: Derive Rules from Data and Encode Data in Weka

Dr. Varun Dutt

Due: Before 11:59 PM on Aug 12, 2026

Readings:

- Class notes and slides from week 1
- Weka website: <http://www.cs.waikato.ac.nz/ml/weka/>

Objectives:

- Derive Classification and Association Rules from dataset.
- Learn how to create an Input file in Weka.

Assignment:

1. For the data shown below:

NAME of training pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
T1	Gabby	Baked	Clogs	Student
T2	Gabby	Roasted	Sandals	Professor
T3	Gabby	Baked	Sandals	Student
T4	Quiet	Fried	Sandals	Professor
T5	Gabby	Fried	Clogs	Student
T6	Quiet	Bakcd	Sandals	Student
T7	Gabby	Fried	Sandals	Professor
T8	Quiet	Fried	Clogs	Student

Please derive the following:

- A. Three classification rules.
- B. Three association rules.

2. Install WEKA and Encode an Input file in WEKA

In this class, we will be using an open-source data-mining tool called WEKA (<http://www.cs.waikato.ac.nz/ml/weka/>).

Please download WEKA and install it on your laptop/computer (the most updated release is WEKA 3.8.1 at the time of writing this assignment).

Now, please read the following pages of the Data-Mining Book: Section 2.4 (Pg. 51 – 56) till Sparse Data. The relevant reading is included as part of this assignment.

Apply the concepts presented to create an .ARFF file for the dataset presented in Question 1 above. Please make sure that the .ARFF file loads correctly in WEKA.

For this purpose, start WEKA 3.6 (with console) -> Explorer -> Open File -> Browse to the directory where the file is located.

Please ensure that the names of all team members are on your submission! Please submit a word/PDF document with your responses to the above questions. In addition, please submit your .ARFF file.